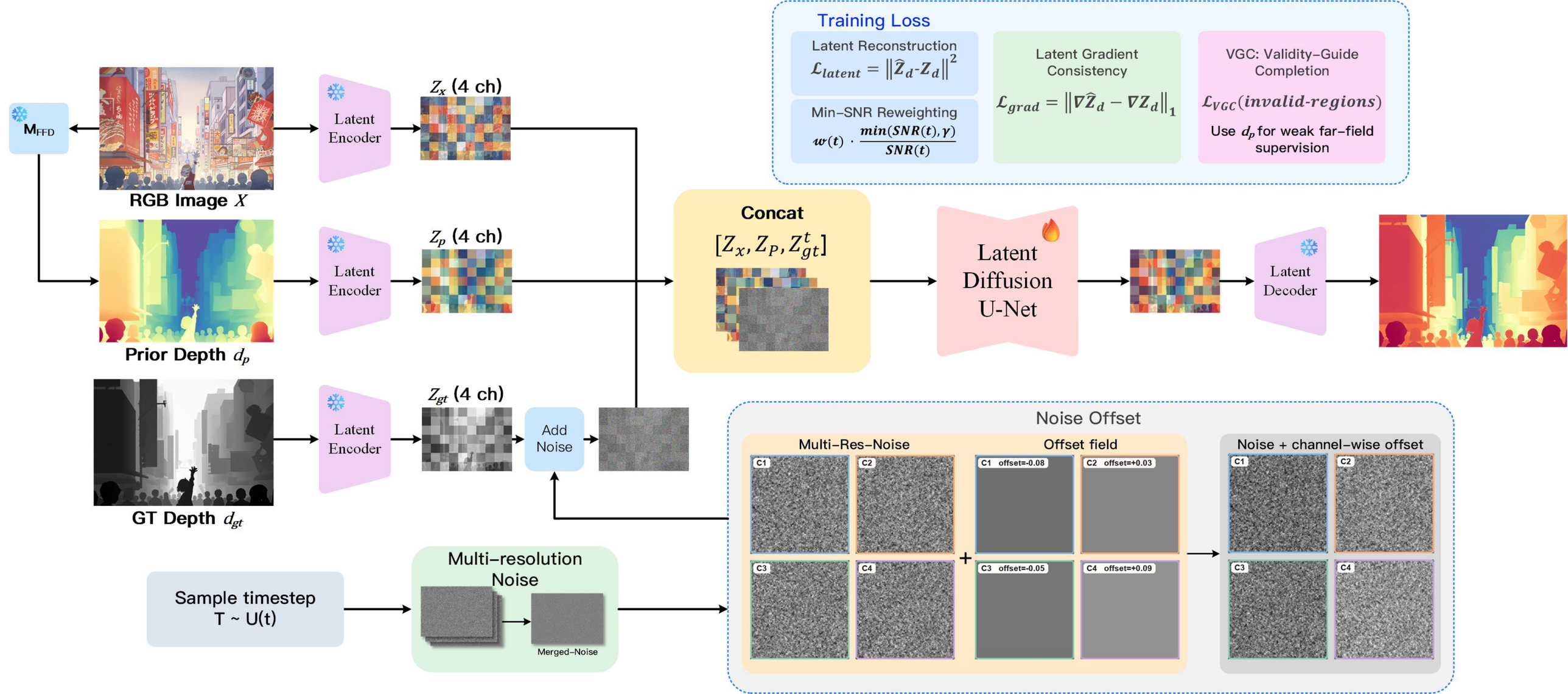

Training Loss
Latent Reconstruction
$\mathcal{L}_{latent} = \|\hat{Z}_d - Z_d\|^2$
Min–SNR Reweighting
$w(t) \cdot \frac{min(SNR(t), \gamma)}{SNR(t)}$
Latent Gradient Consistency
$\mathcal{L}_{grad} = \|\nabla\hat{Z}_d - \nabla Z_d\|_1$
VGC: Validity–Guide Completion
$\mathcal{L}_{VGC}(invalid\text{-}regions)$
Use d_p for weak far–field supervision
M_{FFD}
RGB Image X
Prior Depth d_p
GT Depth d_{gt}
Latent Encoder
Latent Encoder
Latent Encoder
Z_x (4 ch)
Z_p (4 ch)
Z_{gt} (4 ch)
Add Noise
Concat
$[Z_x, Z_P, Z_{gt}^t]$
Latent Diffusion U-Net
Latent Decoder
Noise Offset
Multi–Res–Noise
Offset field
C1
C2
C3
C4
C1 offset=-0.08
C2 offset=+0.03
C3 offset=-0.05
C4 offset=+0.09
Noise + channel–wise offset
Sample timestep
T ~ U(t)
Multi–resolution Noise
Merged–Noise